

Three capability axes vector RAG and knowledge graphs structurally lack

■ SMA

■ dense RAG

Medicine LLM-QA (n = 120 answerable + 120 held-out); LLM and prompt fixed, memory swapped. SMA (teal) vs dense RAG (gray).



Citation

faithful, verifiable provenance
(ALCE-style support)

cite-support score (higher = better)

SMA 0.62

RAG 0.48

Why RAG can't:

retrieved chunks support an answer
only lexically; SMA returns the exact
ontology term + ascension path used

→ a **structural receipt**, not a snippet



Calibrated abstention

knows when it does not know
(grounding AUROC)

known-vs-unknown discrimination

SMA 0.79

RAG 0.55

chance 0.50

Why RAG can't:

cosine similarity stays high even for
unseen diseases, so its gate must
over-abstain (refuses 79% of
answerable to reach recall 0.90).

SMA: recall 0.91 refusing only 45%



Novelty detection

flags out-of-ontology cases
(structural novelty F1)

novelty F1 (higher = better)

SMA 0.79

RAG 0.55

Why RAG falls short:

cosine novelty barely beats chance;
in the pure retrieval suite every
vector baseline scores F1 exactly 0.

SMA: expectation violation = novel